12TH BIOLOGY · CHAPTER 15 · FULL CHAPTER MCQS

Human Nervous System

91 topic-wise MCQs from the complete chapter with answers and reasons

15.1 – 15.2 NERVOUS COORDINATION AND NEURONS

1. Chemical coordination is controlled by the:

(a) Endocrine system (b) Nervous system
(c) Urinary system (d) Digestive system

Answer: (a) Endocrine system — Nervous coordination is controlled by the nervous system.

2. Organs, tissues or cells that detect stimuli are called:

(a) Effectors (b) Neuroglia
(c) Ganglia (d) Receptors

Answer: (d) Receptors — Examples are eyes, ears and skin.

3. Receptors act as:

(a) Effectors (b) Transducers
(c) Insulators (d) Hormones

Answer: (b) Transducers — They change stimuli into nerve impulses.

4. The control centre of nervous coordination is the:

(a) Receptor (b) Effector
(c) Central nervous system (d) Ganglion

Answer: (c) Central nervous system — It consists of the brain and spinal cord.

5. Which of these act as effectors?

(a) Eyes and ears (b) Nerves and ganglia
(c) Skin and tongue (d) Muscles and glands

Answer: (d) Muscles and glands — Muscles contract and glands secrete.

6. The CNS is made of which neurons?

(a) Sensory neurons (b) Inter-neurons
(c) Motor neurons (d) Unipolar neurons

Answer: (b) Inter-neurons — Associative neurons (inter-neurons) make the brain and spinal cord.

7. The structural and functional unit of the nervous system is the:

(a) Neuron (b) Nephron
(c) Nerve (d) Neuroglia

Answer: (a) Neuron — Neurons are also called nerve cells.

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8. Nissl's granules are groups of:

(a) Mitochondria and Golgi bodies
(b) Lysosomes
(c) Ribosomes and rough endoplasmic reticulum
(d) Synaptic vesicles

Answer: (c) Ribosomes and rough endoplasmic reticulum — They are present in the cytoplasm of the cell body.

9. Which part of a neuron carries impulses toward the cell body?

(a) Axon (b) Dendrite
(c) Axon terminal (d) Myelin sheath

Answer: (b) Dendrite — Axons carry impulses away from the cell body.

10. The cells which supply nutrients to neurons, remove wastes and provide immunity are:

(a) Ganglia (b) Nissl's granules
(c) Nociceptors (d) Neuroglia

Answer: (d) Neuroglia — They are also called glial cells.

11. Myelin sheaths in the peripheral nervous system are made by:

(a) Schwann cells (b) Oligodendrocytes
(c) Merkel cells (d) Basal cells

Answer: (a) Schwann cells — Oligodendrocytes make them in the CNS.

12. Myelin sheaths in the central nervous system are made by:

(a) Schwann cells (b) Sensory neurons
(c) Oligodendrocytes (d) Supporting cells

Answer: (c) Oligodendrocytes — Schwann cells make them in the PNS.

13. The non-myelinated gaps in a myelin sheath are called:

(a) Synapses (b) Nodes of Ranvier
(c) Ganglia (d) Ventricles

Answer: (b) Nodes of Ranvier — They occur at intervals of 1 to 2 mm.

14. Clusters of cell bodies of sensory neurons outside the CNS are called:

(a) Nuclei (b) Meninges
(c) Ganglia (d) Papillae

Answer: (c) Ganglia — Ganglia are aggregations of cell bodies outside the CNS.

15. Neurons that conduct impulses from the CNS to effectors are:

(a) Motor neurons (b) Inter-neurons
(c) Sensory neurons (d) Associative neurons

Answer: (a) Motor neurons — Motor neurons are multipolar.

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16. Which neurons occur entirely within the CNS?

(a) Sensory neurons (b) Motor neurons
(c) Unipolar neurons (d) Inter-neurons

Answer: (d) Inter-neurons — They link sensory and motor neurons.

17. The simplest neural pathway is the:

(a) Limbic system (b) Reflex arc
(c) Motor pathway (d) Synapse

Answer: (b) Reflex arc — It controls reflex action.

18. In the knee-jerk reflex, the effector is the:

(a) Quadriceps muscle (b) Hamstring muscle
(c) Patellar tendon (d) Spinal cord

Answer: (a) Quadriceps muscle — It contracts and the lower leg kicks forward.

19. In the knee-jerk reflex, the hamstring motor neurons are inhibited by:

(a) A sensory neuron (b) The brain
(c) An inter-neuron (d) The quadriceps

Answer: (c) An inter-neuron — The inter-neuron releases inhibitory chemicals in the spinal cord.

15.3 – 15.4 NERVE IMPULSE AND SYNAPSE

20. A wave of electrochemical changes that travels along a neuron is called:

(a) Synapse (b) Reflex arc
(c) Nerve impulse (d) Neurotransmitter

Answer: (c) Nerve impulse — This is the definition of nerve impulse.

21. The resting membrane potential of a neuron is about:

(a) +50 mV (b) 0 mV
(c) –120 mV (d) –70 mV

Answer: (d) –70 mV — The inside is negative with respect to the outside.

22. A resting neuron is said to be:

(a) Depolarized (b) Polarized
(c) Hyperpolarized (d) Repolarized

Answer: (b) Polarized — Its inside is negative and outside is positive.

23. The concentration of K^+ inside a neuron is how many times greater than outside?

(a) 30 times (b) 10 times
(c) 3 times (d) 50 times

Answer: (a) 30 times — Na^+ is 10 times greater outside than inside.

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24. For every two K^+ moved in, the sodium–potassium pump moves out:

(a) One Na^+ (b) Two Na^+
(c) Four Na^+ (d) Three Na^+

Answer: (d) Three Na^+ — This makes the inside more negative.

25. Depolarization is caused by the:

(a) Inflow of Cl^- (b) Outflow of K^+
(c) Inflow of Na^+ (d) Outflow of Ca^{2+}

Answer: (c) Inflow of Na^+ — Na^+ gates open and Na^+ rushes in.

26. During depolarization the membrane potential reaches about:

(a) –70 mV (b) +50 mV
(c) 0 mV (d) –50 mV

Answer: (b) +50 mV — It passes zero and reaches +50 mV.

27. Repolarization is caused by the:

(a) Outflow of K^+ (b) Inflow of Na^+
(c) Inflow of Ca^{2+} (d) Inflow of K^+

Answer: (a) Outflow of K^+ — Na^+ gates close and K^+ gates open.

28. The period in which a neuron cannot generate another action potential is the:

(a) Resting period (b) Latent period
(c) Refractory period (d) Active period

Answer: (c) Refractory period — It occurs during hyperpolarization.

29. Hyperpolarization is due to a slight delay in the closing of:

(a) Na^+ gates (b) K^+ gates
(c) Ca^{2+} gates (d) Cl^- gates

Answer: (b) K^+ gates — Extra K^+ leaves the cell.

30. An action potential on a region of the neuron lasts for about:

(a) Ten seconds (b) One second
(c) One minute (d) One millisecond

Answer: (d) One millisecond — It is very brief.

31. The velocity of a nerve impulse is greater if the axon is:

(a) Large in diameter or myelinated (b) Thin and non-myelinated
(c) Short (d) Branched

Answer: (a) Large in diameter or myelinated — The strength of the action potential remains constant.

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32. Myelinated neurons conduct impulses at speeds of up to:

(a) 3 m/s (b) 120 m/s
(c) 50 m/s (d) 12 m/s

Answer: (b) 120 m/s — Non-myelinated neurons conduct at 1 to 3 m/s.

33. Saltatory conduction is up to _____ faster than conduction in the fastest non-myelinated axons.

(a) 5 times (b) 10 times
(c) 50 times (d) 120 times

Answer: (c) 50 times — The impulse jumps from node to node.

34. The fluid-filled gap between two cells at a synapse is the:

(a) Synaptic cleft (b) Synaptic knob
(c) Synaptic vesicle (d) Node of Ranvier

Answer: (a) Synaptic cleft — The two cells are not in direct contact.

35. Neurotransmitters are stored in:

(a) Dendrites (b) Nissl’s granules
(c) Nodes of Ranvier (d) Synaptic vesicles

Answer: (d) Synaptic vesicles — They lie in the synaptic knobs.

36. Neurotransmitters are released from the presynaptic membrane by:

(a) Exocytosis (b) Endocytosis
(c) Osmosis (d) Active transport

Answer: (a) Exocytosis — Vesicles fuse with the membrane.

37. The synapse between a motor neuron and a muscle fibre is called:

(a) Reflex arc (b) Corpus callosum
(c) Ganglion (d) Neuromuscular junction

Answer: (d) Neuromuscular junction — Acetylcholine acts here.

38. Which is an inhibitory neurotransmitter?

(a) Acetylcholine (b) GABA
(c) Glutamate (d) Norepinephrine

Answer: (b) GABA — GABA is the chief inhibitory neurotransmitter of the CNS.

39. Inhibitory neurotransmitters open channels for:

(a) Sodium ions (b) Calcium ions
(c) Chloride ions (d) Potassium ions

Answer: (c) Chloride ions — Chloride ions make the inside more negative.

40. About 90% of serotonin is produced in the:

(a) Brain (b) Spinal cord
(c) Adrenal glands (d) Intestine

Answer: (d) Intestine — About 10% is produced in the CNS.

41. Which neurotransmitters block pain signals?

(a) Glutamate (b) Endorphins
(c) Acetylcholine (d) Epinephrine

Answer: (b) Endorphins — They reduce the release of pain-transmitting neurotransmitters.

42. The neurotransmitter that controls body movements, pleasure and motivation is:

(a) Glycine (b) GABA
(c) Dopamine (d) Acetylcholine

Answer: (c) Dopamine — It is a biogenic amine made in the brain.

15.5 BASIC ORGANIZATION OF HUMAN NERVOUS SYSTEM

43. The outermost thick layer of the meninges is the:

(a) Pia mater (b) Arachnoid
(c) Cortex (d) Dura mater

Answer: (d) Dura mater — It contains larger blood vessels.

44. The largest portion of the human brain is the:

(a) Cerebrum (b) Cerebellum
(c) Medulla oblongata (d) Thalamus

Answer: (a) Cerebrum — It is divided into two cerebral hemispheres.

45. The two cerebral hemispheres are connected by the:

(a) Pons (b) Hippocampus
(c) Corpus callosum (d) Midbrain

Answer: (c) Corpus callosum — It is a band of axons.

46. Vision is controlled by the:

(a) Frontal lobes (b) Occipital lobes
(c) Temporal lobes (d) Parietal lobes

Answer: (b) Occipital lobes — The visual cortex lies in the occipital lobe.

47. Planning, reasoning and abstract thought are functions of the:

(a) Frontal lobes (b) Occipital lobes
(c) Cerebellum (d) Pons

Answer: (a) Frontal lobes — These are executive functions.

48. The auditory cortex is located in the:

(a) Frontal lobe (b) Temporal lobe
(c) Parietal lobe (d) Occipital lobe

Answer: (b) Temporal lobe — It is a sensory area.

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49. The cerebral cortex is made of:

(a) White matter (b) Myelinated axons only
(c) Grey matter (d) Meninges

Answer: (c) Grey matter — It contains cell bodies and non-myelinated axons.

50. The control centre for heart rate and respiration rate is the:

(a) Cerebellum (b) Thalamus
(c) Hippocampus (d) Medulla oblongata

Answer: (d) Medulla oblongata — It is a part of the brainstem.

51. Which part transmits impulses between the cerebral hemispheres and the cerebellum?

(a) Pons (b) Midbrain
(c) Thalamus (d) Amygdala

Answer: (a) Pons — It is a part of the brainstem.

52. Body temperature, hunger and thirst are regulated by the:

(a) Thalamus (b) Pons
(c) Cerebellum (d) Hypothalamus

Answer: (d) Hypothalamus — It also controls the pituitary gland.

53. Short-term memory is consolidated into long-term memory by the:

(a) Amygdala (b) Hippocampus
(c) Thalamus (d) Medulla

Answer: (b) Hippocampus — It is a part of the limbic system.

54. Posture and coordination of muscle actions are controlled by the:

(a) Cerebrum (b) Midbrain
(c) Cerebellum (d) Hypothalamus

Answer: (c) Cerebellum — It lies below and behind the cerebral hemispheres.

55. How many ventricles are present in the human brain?

(a) Four (b) Three
(c) Two (d) Five

Answer: (a) Four — Two lateral, the third and the fourth ventricle.

56. In the spinal cord, grey matter is:

(a) On the outer side (b) In the meninges
(c) Absent (d) In the inner H-shaped zone

Answer: (d) In the inner H-shaped zone — White matter makes the outer zone.

57. The number of pairs of cranial nerves in humans is:

(a) 10 (b) 12
(c) 31 (d) 33

Answer: (b) 12 — Two pairs arise from the cerebrum and ten from the brainstem.

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58. The number of pairs of spinal nerves in humans is:

(a) 12 (b) 21
(c) 31 (d) 43

Answer: (c) 31 — All of them are mixed nerves.

59. The cranial nerve for the sense of smell is nerve:

(a) II (b) I
(c) VIII (d) X

Answer: (b) I — It is the olfactory nerve.

60. How many pairs of thoracic spinal nerves are there?

(a) 5 (b) 8
(c) 1 (d) 12

Answer: (d) 12 — 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal.

61. The ventral root of a spinal nerve contains axons of:

(a) Motor neurons (b) Sensory neurons
(c) Inter-neurons (d) Glial cells

Answer: (a) Motor neurons — The dorsal root contains sensory neurons.

62. The somatic nervous system controls:

(a) Smooth muscles (b) Glands
(c) Skeletal muscles (d) Heart muscle

Answer: (c) Skeletal muscles — It is voluntary.

63. Motor neurons of the sympathetic system originate from the:

(a) Brain and sacral region (b) Cerebellum
(c) Cervical region only (d) Thoracic and lumbar regions

Answer: (d) Thoracic and lumbar regions — Parasympathetic neurons arise from the brain and sacral region.

64. Many parasympathetic neurons are present in which cranial nerve?

(a) Optic (b) Vagus
(c) Olfactory (d) Facial

Answer: (b) Vagus — The vagus is the tenth cranial nerve.

65. The sympathetic system causes the pupil of the eye to:

(a) Constrict (b) Close
(c) Dilate (d) Remain unchanged

Answer: (c) Dilate — The parasympathetic system constricts it.

66. The parasympathetic system makes the urinary bladder:

(a) Contract (b) Relax
(c) Dilate (d) Stop working

Answer: (a) Contract — The sympathetic system relaxes it.

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15.6 – 15.9 RECEPTORS, DRUGS, DISORDERS AND DIAGNOSTIC TESTS

67. The number of taste buds in humans is about:

(a) 1,000 (b) 10,000
(c) 100,000 (d) One million

Answer: (b) 10,000 — Most lie between the papillae of the tongue.

68. Taste buds for sweet taste are concentrated at the _____ of the tongue.

(a) Centre (b) Sides
(c) Back (d) Tip

Answer: (d) Tip — Sour at the sides and bitter at the back.

69. Taste impulses travel along cranial nerves:

(a) I, II and III (b) IV, V and VI
(c) VII, IX and X (d) XI and XII

Answer: (c) VII, IX and X — They go to the brainstem and then to the cerebral cortex.

70. The perception of smell is produced by the:

(a) Olfactory bulb (b) Taste pore
(c) Thalamus (d) Cerebellum

Answer: (a) Olfactory bulb — It receives impulses through cranial nerve I.

71. In the nasal epithelium, old sensory neurons are replaced by:

(a) Supporting cells (b) Merkel cells
(c) Basal cells (d) Schwann cells

Answer: (c) Basal cells — Basal cells act as stem cells.

72. Light touch and vibration are detected by:

(a) Pacinian corpuscles (b) Meissner’s corpuscles
(c) Ruffini endings (d) Nociceptors

Answer: (b) Meissner’s corpuscles — They lie in the upper layer of skin.

73. Sustained pressure and stretching are detected by:

(a) Merkel cells (b) Hair follicle receptors
(c) Meissner’s corpuscles (d) Ruffini endings

Answer: (d) Ruffini endings — They are located in the dermis.

74. Touch receptors are:

(a) Mechanoreceptors (b) Chemoreceptors
(c) Photoreceptors (d) Nociceptors

Answer: (a) Mechanoreceptors — They respond to physical stimuli.

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75. Which of these is a narcotic?

(a) Nicotine (b) Caffeine
(c) Codeine (d) Alcohol

Answer: (c) Codeine — Narcotics include morphine, codeine and oxycodone.

76. The condition in which the body becomes less sensitive to a drug is:

(a) Tolerance (b) Dependence
(c) Addiction (d) Withdrawal

Answer: (a) Tolerance — Higher doses are needed for the same response.

77. Addictive drugs cause presynaptic neurons to release:

(a) Adenosine (b) GABA
(c) Glycine (d) Dopamine

Answer: (d) Dopamine — Dopamine gives feelings of pleasure and reward.

78. In tolerance, the brain reduces the number of:

(a) Synaptic vesicles
(b) Dopamine receptors on the postsynaptic neuron
(c) Neurons
(d) Calcium gates

Answer: (b) Dopamine receptors on the postsynaptic neuron — So higher doses are needed.

79. Cannabis is obtained from:

(a) Indian hemp (b) Opium poppy
(c) Tobacco (d) Coffee plant

Answer: (a) Indian hemp — Its scientific name is *Cannabis sativa*.

80. Heroin is derived from the:

(a) Tobacco plant (b) Opium poppy plant
(c) Indian hemp (d) Tea plant

Answer: (b) Opium poppy plant — It is a highly addictive opioid.

81. Stroke is an example of a _____ disorder.

(a) Degenerative (b) Infectious
(c) Functional (d) Vascular

Answer: (d) Vascular — Blood supply to a part of the brain is interrupted.

82. Inability to express through words is called:

(a) Dementia (b) Migraine
(c) Aphasia (d) Seizure

Answer: (c) Aphasia — It is a common symptom of stroke.

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83. Inflammation of the meninges due to infection is:

(a) Stroke (b) Meningitis
(c) Migraine (d) Multiple sclerosis

Answer: (b) Meningitis — Bacterial meningitis is the most severe.

84. Brain tumour is a _____ disorder.

(a) Structural (b) Functional
(c) Vascular (d) Infectious

Answer: (a) Structural — It damages the physical structure of the nervous system.

85. Headache is an example of a _____ disorder.

(a) Structural (b) Vascular
(c) Degenerative (d) Functional

Answer: (d) Functional — There is no apparent structural cause.

86. The disease in which the myelin sheath is damaged is:

(a) Parkinson's disease (b) Huntington's disease
(c) Multiple sclerosis (d) Meningitis

Answer: (c) Multiple sclerosis — It is an autoimmune disease of the CNS.

87. In Parkinson's disease, the neurons that die produce:

(a) Serotonin (b) Dopamine
(c) GABA (d) Acetylcholine

Answer: (b) Dopamine — This leads to movement issues.

88. EEG records the _____ of the brain.

(a) Electrical activity (b) Blood flow
(c) Temperature (d) Hydrogen atoms

Answer: (a) Electrical activity — Electrodes are attached to the scalp.

89. CT scan uses:

(a) Radio waves (b) Magnets
(c) Ultrasound (d) X-rays

Answer: (d) X-rays — Different tissues absorb X-rays to different degrees.

90. MRI is based on the magnetic properties of:

(a) Oxygen atoms (b) Carbon atoms
(c) Hydrogen atoms (d) Calcium ions

Answer: (c) Hydrogen atoms — Different tissues have different amounts of hydrogen atoms.

91. The doughnut-shaped machine of a CT scan is called a:

(a) Scanner (b) Gantry

(c) Amplifier (d) Dialyzer

Answer: (b) Gantry — It contains a rotating X-ray tube.

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